## Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the goal of linear regression? — 1 / 1 point

- (●) Create a linear model that minimizes the sum of squared residuals
- ( ) Create a nonlinear model that minimizes the sum of squared residuals
- ( ) Create a linear model that maximizes the sum of squared residuals

**Correct**
Correct! The goal of linear regression is to create a linear model that minimizes the sum of squared residuals.

2. Which of the following are assumptions of linear regression (select all that apply)? — 1 / 1 point

- [x] Homoscedasticity

**Correct**
Correct! Homoscedasticity states that the variance of the residual errors (the errors between the predicted and actual values) is constant across all values of the independent variables. If this assumption is violated (heteroscedasticity), the model may be inefficient or biased.

- [x] Normality

**Correct**
Correct! The residual errors must follow a normal distribution. This assumption is especially important for hypothesis testing and confidence interval estimation.

- [x] Independence of observations

**Correct**
Correct! The observations in the dataset must be independent of each other. There should be no correlation between the residual errors of the observations.

3. What does Least absolute shrinkage and selection operator (Lasso) introduce to a linear regression model? — 1 / 1 point

- ( ) Homoscedasticity
- (●) Sparsity
- ( ) Multicollinearity

**Correct**
Correct! Lasso performs feature selection and regularization of the selected feature weights when applied to a linear regression model, making it more sparse and thus more interpretable.

4. For classification problems, which model is preferred? — 1 / 1 point

- ( ) Linear Regression
- (●) Logistic Regression
- ( ) Poisson Regression

**Correct**
Correct! For classification problems, we would prefer to work with probabilities between 0 and 1, so we can use the logistic function to map this.

5. Why is the interpretation of logistic regression more challenging than linear regression? — 1 / 1 point

- ( ) The interpretation of the weights is subtractive and not additive
- ( ) The interpretation of the weights is additive and not subtractive
- (●) The interpretation of the weights is multiplicative and not additive

**Correct**
Correct! The interpretation of logistic regression is more difficult than linear regression because the interpretation of the weights is multiplicative and not additive.

6. What model should I use if my target outcome does not follow a Gaussian distribution? — 1 / 1 point

- (●) Generalized Linear Model (GLM)
- ( ) Generalized Additive Model (GAM)
- ( ) Generalized Multiplicative Model (GMM)

**Correct**
Correct! If your target outcome does not follow a Gaussian distribution, you should use a GLM, which allows non-Gaussian outcome distributions.

7. What model should I use if the true relationship between the features and outcomes is NOT linear? — 1 / 1 point

- ( ) Generalized Linear Model (GLM)
- (●) Generalized Additive Model (GAM)
- ( ) Generalized Multiplicative Model (GMM)

**Correct**
Correct! If the true relationship between the features and outcomes is NOT linear, you should use a GAM, which gives the option to allow nonlinear relationships between some features and the output.

8. What is the link function for Logistic Regression? — 1 / 1 point

- (●) Logit function
- ( ) Bernoulli distribution
- ( ) Poisson distribution

**Correct**
Correct! Logistic Regression assumes a Bernoulli distribution (0 or 1) for the outcome and links the expected mean and the weighted sum using the logit function.

9. How do GAMs learn nonlinear functions? — 1 / 1 point

- ( ) Logit function
- ( ) Quadratic function
- (●) C: Spline functions

**Correct**
Correct! Splines are used to approximate other, more complex functions and are used in GAMs to learn nonlinear functions.

10. What is a significant con with using GLMs or GAMs? — 1 / 1 point

- ( ) Modifications of the linear model are not based in solid statistical theory
- ( ) They are not well documented
- (●) C: Most modifications of the linear model make the model less interpretable

**Correct**
Correct! Most modifications of the linear model make the model less interpretable. The link function in GLMs, for example, complicates interpretation of the result.